

# Robust Choices That Never Settle

An exact finite–infinite puzzle about resilience, choice, and randomness

Research article prepared for Vladimir Reshetnikov

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## Abstract

A committee must choose one of two alternatives, treat their names symmetrically, and withstand adversarial changes to its votes. How robust can a finite rule be, and can increasingly robust finite rules settle into a neutral infinite rule? We solve the finite optimization exactly by an antipodal-core reduction to Kleitman’s diameter theorem: odd majority simultaneously maximizes every Hamming robustness probability, and an even committee has the same optimum as a committee with one fewer member. We then prove a sharp finite-prefix inequality converting adversarial vulnerability into conditional predictability. Its optimal constant is the reciprocal of a Hamming-ball volume fraction. The inequality shows that even vanishing one-vote vulnerability forces the decisions to disagree asymptotically half the time with every fixed measurable candidate limit. They oscillate on a conull, comeager set.

For stronger resilience, we extract a deterministic sequence of committee sizes whose decision map is a continuous open surjection with output law arbitrarily close, in a bounded-density sense, to fair Bernoulli measure. Along these sizes, robustness tends to infinity and decisions are statistically normal almost surely, whereas generically the frequencies swing from zero to one and one-vote fragility recurs. An everywhere neutral infinite rule nevertheless exists under choice, with maximally nonmeasurable fibres. For biased product laws, completed-measure existence has an exact square-summability threshold. The article includes complete proofs of the derived statements, explicit separations, an exact finite verifier, a ProveIt integration map, and further research questions. Classical ingredients and unverified priority claims are carefully distinguished.

**Keywords.** Boolean functions; Hamming robustness; Kleitman diameter theorem; mixing convergence; finite-change equivalence; Baire category; choice.

[write] **Status and trust boundary, added 5 October 2026, batch 114.** This is an unrefereed external manuscript, filed in ProveIt as the single-source research report robust-neutral-choice (provenance, the sources as the write read them, relation to the repository, the collected non-claims and the reading conventions are in Section 1.4). Its author line and its PDF author field read “Research article prepared for Vladimir Reshetnikov”; the package says nothing about how the text was produced, and the write adds no authorship statement. No statement of the report is formalized in Lean or Rocq (Remark 10.1 says which proved Lean declarations bear on the category half of Theorem 8.1). *What rests on what.* The finite optimum, Theorem 2.1, is a corollary of Kleitman’s diameter theorem (Theorem 2.3, quoted without proof and credited, as the source credits it, to [4, 5]); the source proves the antipodal-core reduction (Lemma 2.2) and the matching majority construction, and claims no novelty for the optimum. Theorem 9.1 takes Kakutani’s product-measure dichotomy [9] as an explicit input. Every other theorem, lemma and proposition of the source is proved in the text from standard probability, measure theory and topology (the central limit theorem and Stirling’s formula in Proposition 3.1; conditional expectations, Borel–Cantelli, Radon–Nikodym and compactness later); the mixing conclusions are consequences of Rényi’s theory of mixing sequences [6] and the sparse extraction belongs to the subsequence-principle literature [7, 8], both credited by the source. The exhaustive program code/verify.py checks the finite statements in dimensions one to five and proves nothing in higher dimension. The source claims no novelty or priority and does not claim to settle a published open

problem. The write adds Remark [10.1](#), Proposition [11.1](#) (the regularity obstruction for more than two labels), short notes marked [write] and Section [11.1](#).

## Contents

|           |                                                                         |           |
|-----------|-------------------------------------------------------------------------|-----------|
| <b>1</b>  | <b>The problem and its mathematical setting</b>                         | <b>4</b>  |
| 1.1       | Why this problem fits the requested intersection . . . . .              | 4         |
| 1.2       | Definitions and conventions . . . . .                                   | 4         |
| 1.3       | What is classical, what is proved here, and what remains open . . . . . | 5         |
| 1.4       | Provenance, sources and reading conventions . . . . .                   | 5         |
| <b>2</b>  | <b>An exact solution for every finite committee</b>                     | <b>8</b>  |
| 2.1       | The antipodal-core reduction . . . . .                                  | 9         |
| 2.2       | Examples and the distinction from random-noise stability . . . . .      | 10        |
| <b>3</b>  | <b>The scale of finite resilience</b>                                   | <b>10</b> |
| <b>4</b>  | <b>A sharp inequality for what early voters can predict</b>             | <b>12</b> |
| 4.1       | The smallest vulnerable region in a nonconstant cube . . . . .          | 12        |
| 4.2       | Complementary fibres cancel . . . . .                                   | 13        |
| 4.3       | Why the constant cannot be improved . . . . .                           | 13        |
| 4.4       | Conditional probabilities and comparisons between committees . . . . .  | 13        |
| <b>5</b>  | <b>Why increasingly resilient decisions cannot settle</b>               | <b>14</b> |
| 5.1       | An even weaker qualitative hypothesis . . . . .                         | 15        |
| 5.2       | One-vote resilience does not imply two-vote resilience . . . . .        | 15        |
| <b>6</b>  | <b>Extracting a continuous open stream of almost fair decisions</b>     | <b>16</b> |
| 6.1       | The deterministic selection and the likelihood bounds . . . . .         | 16        |
| 6.2       | Why the decision map is open and onto . . . . .                         | 17        |
| 6.3       | A coupling interpretation . . . . .                                     | 17        |
| 6.4       | An explicit schedule for ordinary majorities . . . . .                  | 17        |
| <b>7</b>  | <b>Opposite answers from probability and category</b>                   | <b>17</b> |
| 7.1       | Why no fair-frequency conclusion holds for all horizons . . . . .       | 19        |
| <b>8</b>  | <b>A perfect infinite rule exists under choice</b>                      | <b>19</b> |
| <b>9</b>  | <b>How a small statistical bias changes the infinite answer</b>         | <b>20</b> |
| 9.1       | The product-measure input . . . . .                                     | 21        |
| 9.2       | An explicit measurable part of the biased rule . . . . .                | 21        |
| <b>10</b> | <b>Computation, reproducibility, and the ProveIt connection</b>         | <b>22</b> |
| 10.1      | What the supplied computation checks . . . . .                          | 22        |
| 10.2      | The source snapshot and what is already formalized . . . . .            | 22        |
| 10.3      | A practical formalization order . . . . .                               | 24        |
| <b>11</b> | <b>Further research questions and topics</b>                            | <b>24</b> |
| 11.1      | Further questions and research . . . . .                                | 25        |
| <b>A</b>  | <b>A concise problem statement suitable for sharing</b>                 | <b>29</b> |
| <b>B</b>  | <b>Scope and assumptions at a glance</b>                                | <b>29</b> |

# 1 The problem and its mathematical setting

Imagine a committee receiving a binary vote from each member. A rule must always return a decision. Renaming the two alternatives must rename the decision too. After the votes are cast, an adversary sees the entire configuration and may change up to  $r$  votes. We ask how often the original decision survives *every* such intervention. The votes are fair and independent only for measuring the fraction of robust inputs; the intervention itself is worst case.

Now let a single infinite stream of votes feed committees of increasing size. Suppose each fixed intervention budget becomes harmless with probability tending to one. It is tempting to expect an eventually settled decision, insensitive to finite alterations of the stream. The principal surprise is that the two requirements pull in different directions: improving resilience can erase the influence of every fixed initial segment without producing a measurable limit.

**Problem 1.1** (Robust neutral choice). Answer the following linked questions.

- (i) For  $n$  fair binary inputs, what is the largest fraction on which a neutral deterministic rule withstands all changes to at most  $r$  coordinates? Can one rule attain the optimum for every  $r$ ?
- (ii) Can neutral rules on growing prefixes become resistant to every fixed number of changes and nevertheless converge on a substantial set of streams?
- (iii) If convergence is impossible, how much randomness and how much topological freedom remain in the sequence of decisions?
- (iv) Can a perfectly robust neutral rule be defined directly on infinite streams? How do choice and measurability affect the answer?

## 1.1 Why this problem fits the requested intersection

Epoch AI’s public biographical note describes Elliot Glazer’s interests as set theory and formal systems, choice paradoxes, proof-assistant foundations, and puzzles with finite and infinite versions [1]. His dissertation *Interactions Among Principles of Choice, Forcing, and Measure Theory* examines the relation between choice and regularity, including transversals of Borel equivalence relations [2]. These sources support the proposed fit; the problem has not been reviewed or endorsed by him.

ProveIt provides a concrete starting point. Its `GenericErgodicity.lean` develops finite-change generic ergodicity and obstructions to regular finite-label rules. The abstract theorems `generic_const` and `fixed_label`, together with `Cantor.transitive`, supply the category mechanism used here [3]. The present question adds exact finite robustness and a quantitative analysis of what finite approximations can do. Section 10 identifies the precise existing ingredients and the additional formalization tasks.

## 1.2 Definitions and conventions

Write  $Q_n = \{-1, 1\}^n$ , with uniform probability  $\mu_n$ , and let  $d_H$  denote Hamming distance. Complementation is  $x \mapsto -x$ .

**Definition 1.2.** A rule  $f : Q_n \rightarrow \{-1, 1\}$  is *neutral*, or self-dual, if

$$f(-x) = -f(x) \quad (x \in Q_n). \quad (1)$$

Its *change distance* is

$$R_f(x) = \min\{d_H(x, y) : f(y) \neq f(x)\}. \quad (2)$$

For an integer  $r \geq 0$ , define

$$\rho_f(r) = \mu_n(R_f > r), \quad \varepsilon_f(r) = 1 - \rho_f(r) = \mu_n(R_f \leq r). \quad (3)$$

Thus  $R_f \geq 1$ , and  $R_f = 1$  means that a single vote can change the answer.

Neutrality implies that  $f$  is balanced: its two fibres have equal size. It does not require invariance under permutations of voters. We use the term *neutral*, not *anonymous*, for this reason. For odd committees, the optimal majority rule is anonymous as well. For an even committee, a neutral anonymous deterministic rule cannot exist: a tied configuration and its complement are permutations of one another, but neutrality demands opposite outputs.

On  $\Omega = \{-1, 1\}^{\mathbb{N}}$ , let  $\mu$  be fair product measure and let  $\mathcal{F}_m = \sigma(X_1, \dots, X_m)$ . A finite rule is also viewed as a function on  $\Omega$  by ignoring coordinates after its horizon. We write  $x E_0 y$  when the two streams differ at only finitely many positions. All measure-theoretic statements allow the completion of the displayed product measure. “Comeager” means that the complement is a countable union of nowhere dense sets in the usual Cantor topology.

### 1.3 What is classical, what is proved here, and what remains open

This is a designed problem with a rigorous solution, rather than a claim to have settled a previously advertised major conjecture. The finite extremum is a corollary of Kleitman’s classical theorem [4, 5]. The nonconvergence mechanism belongs to Rényi’s theory of mixing sequences [6]; sparse extraction belongs to the wider subsequence-principle literature [7, 8]. We give direct proofs specialized to the puzzle.

The sharp prefix inequality, its matching examples, the quantitative open recoding theorem, and the assembled probability/category consequences are proved below. Their proofs do not certify historical priority. The two substantial theorems quoted without proof are the explicitly stated diameter theorem and, in the optional biased extension, Kakutani’s product-measure dichotomy. No new Lean verification is claimed. The further questions in Section 11 are research proposals, with their status qualified there.

### 1.4 Provenance, sources and reading conventions

[write] **Added 5 October 2026, batch 114.** *Provenance.* The report is built from one manuscript, the archive `robust_choices.zip` (617,494 bytes; wrapper directory `robust_choices/`, 12 files), which reached ProveIt’s docs/incoming in the arrival commit 2399df2bd (5 October 2026, “New research reports”), outside the session bundle of the intake, and was placed as batch 114 in 99053b5d1. The text is the delivered `robust_choices.tex` (1,539 lines), shipped as `article.tex`; the delivered code, data and figure files are shipped unchanged (the README lists them). The delivered 22-page PDF is not shipped; it and the delivery README (staged at placement and replaced by the report README) are retrievable from the arrival commit. The package pins ProveIt revision 3b9458b31f459b480ab43a068d7cc56f4cfdabb28 (Section 10, bibliography entry [3] and the delivery README); that commit is the batch-98D placement commit of this repository and an ancestor of the commit this write is based on. The author line and the PDF author field read “Research article prepared for Vladimir Reshetnikov”. The package contains no statement of how the text was produced, no “unrefereed” or “prepared for private review” line, no checksum manifest and no personal data. Single source: no merge choice was made. The write prefixed the 79 delivered labels with `rch:` and updated every reference; it added the notes marked [write], this subsection, Remark 10.1, Section 11.1 with Proposition 11.1, and nothing else. No statement, proof or number of the source was changed: the added remark and proposition are the only numbered statements of their sections and the added displays are unnumbered, so every delivered theorem and equation keeps its number.

*The sources, as the write read them* (5 October 2026).

- Huang, Klurman and Pohoata [5], arXiv version 1: their Theorem 1.1 is Theorem 2.3 as printed here (both parities, diameter at most  $D < n$ , both bounds sharp, the same extremal sets), credited to Kleitman as the solution of a conjecture of Erdős; the paper gives a new, algebraic proof. Read.
- Kleitman [4]: the bibliographic data were confirmed through the DOI; the paper itself was not read.
- Wu, Li, Feng, Liu and Yu [10]: abstract of version 2 (10 June 2025) read. It states Kleitman’s theorem with the odd bound in the form  $\sum_{i \leq s} \binom{n}{i} + \binom{n-1}{s}$ , credits Frankl (2017) with the characterization of the extremal families and a first stability result, and proves a second one; it concerns extremal families of a given diameter, as Question Q2 says. The body was not read.
- Kakutani [9], Rényi [6], Aldous [7], Aldous and Eagleson [8]: not read by the write. The statement used in Section 9, equivalence of the two product measures exactly when the product (50) is positive, is the usual form of Kakutani’s dichotomy for these products.
- Epoch AI’s biographical note [1] and Glazer’s dissertation [2]: motivation only, not read by the write. The source says that the problem has not been reviewed or endorsed by Glazer.
- The repository at the pin and at the write’s base commit: see the note in Section 10.

*What was checked at intake.* The batch-114 dossier read the manuscript in full and checked the diameter bound  $n - 2r - 1$  of the core and both parities of the reduction to Kleitman’s theorem, the nine-voter row, the induction of Lemma 4.2, the fibre cancellation, the sharpness sandwich (23), the code construction (complement invariance needs  $\sum_{v \neq 0} v = 0$ , true for  $d \geq 2$ ), the likelihood product bounds, openness by nested compactness, the singular case  $A' = A \setminus \kappa A$  of Theorem 9.1 and the example  $\gamma \leq 1/2$ ; no error. Independent programs confirmed the nine-voter row exactly and brute-forced the construction of Proposition 5.3 for  $d = 2$ ,  $N = 3$  with  $L = 1$  and  $L = 3$ : the rule is neutral,  $R_f \leq 2$  everywhere, and  $\varepsilon_f(1) = u_N + t_L - u_N t_L$  exactly. The delivered verifier passed on a copy, and its four outputs equal the shipped data files up to newline style (the CSV byte for byte). The write reread every proof, found no error, and made the two exhaustive computations reported in Section 11.1.

*Relation to the repository.*

- *Formal status.* No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. The Lean file the source cites, `SetTheory/Cardinals/Cardinals/Countable/GenericErgodicity.lean`, proves the topological zero-one law and its two corollaries that the category half of Theorem 8.1 needs; Remark 10.1 assembles them and names what is missing.
- The large-cardinal synthesis (`SetTheory/Cardinals/docs/cardinals/research-synthesis/`), Proposition 15.3, prop:baire (the Baire-measurable classification of finite-label rules at  $\lambda = \omega$ ), is what that Lean file formalizes; its discussion notes that a selection from a quotient of  $E_0$  type cannot be Baire measurable, the mechanism of the category half of Theorem 8.1. There the symmetries are permutations of coordinates; here the symmetry is global complementation.
- noisy-parity-cubes (same collection and batch; *see also*): a total parity label there changes sign under every *single* flip and is nonmeasurable for fair measure (np:found:choice, np:found:regularity); a rule here is *invariant* under finite changes and changes sign under global complementation. The biased thresholds differ accordingly: every parity label is measurable for the product of Bernoulli( $q_i$ ) laws exactly when  $\sum_i \min(q_i, 1 - q_i) < \infty$  (np:found:biased), whereas a measurable rule of the present kind exists exactly when  $\sum_i (p_i - \frac{1}{2})^2 = \infty$  (Theorem 9.1). Neither source cites the other; no theorem is shared.
- Batch 114 also filed freiling-symmetry-blacklists, bounded-width-power-set-compression and random-bits-arithmetic-cuts in the same category, and measurable-box-games shares the motivation from Glazer’s work; no theorem is shared with any of them.
- *Stale claims.* The source’s three statements about the repository (what GenericErgodicity.lean contains, the role of FiniteCycles.lean, and that targeted searches did not find the quantitative development) are still correct at the write’s base commit (note in Section 10); nothing needed correcting.

[write] **What is not claimed, collected from the source.** The problem is designed; the source does not claim to have settled “a previously advertised major conjecture”, its proofs “do not certify historical priority”, and its delivery README says that it “makes no unverified claim of worldwide novelty or a breakthrough on a published open problem”. The finite optimum is a corollary of Kleitman’s theorem, quoted without proof; Theorem 9.1 is “a deduction from a classical product-measure theorem, not a new version of Kakutani’s theorem”, concerns existence in one completed biased law and gives no universally measurable neutral selector. The mixing mechanism is Rényi’s and sparse extraction belongs to the subsequence-principle literature; the source gives direct proofs specialized to the puzzle. No new Lean or Rocq verification is claimed; the exact finite program is a check, not a proof of the infinite-dimensional theorems, and its tables for larger  $n$  evaluate the formula rather than enumerate rules. Equation (5) is not a claim about every fairness criterion or voting model; the attaining rules are not classified. The constant of Theorem 4.1 is sharp only uniformly over dimensions, not for each fixed  $(n, m, r)$ . In (28) the candidate  $G$  is fixed before the limit; nothing is uniform over  $G$  depending on  $n$ , and selecting a constant-sign subsequence after seeing the input is not excluded. Proposition 5.2 says nothing about small coalitions. The selected outputs of Theorem 6.1 need not be independent (equivalence with product measure is the claim); the coupling lives on an enlarged probability space and does not provide an independent sequence as a function of the input. (44) is a homomorphism of equivalence relations on  $E$ , not a reduction, and  $H|_E$  is not claimed open or onto. The choice construction does not identify the existence of the rule with full choice, and its exact weak-choice strength is left open. The Lean declarations should not be reported as “an already named, directly matching repository theorem”; the repository searches are “evidence about the inspected material, not a certification of novelty”. The explicit majority schedule is not proposed as optimal. The research questions are proposals; a question labelled a target is not asserted to be a previously published open problem. Randomized rules must be neutral for each fixed seed. The problem has not been reviewed or endorsed by Glazer.

[write] **Reading conventions.** The abstract’s “For stronger resilience, we extract” covers more than it needs: the open map and the bounded density of Theorem 6.1 need only one-vote resilience (26); resilience at every fixed budget (27) is the hypothesis of (36) and of Theorem 7.1. The cube is  $\{-1, 1\}^n$  throughout the text, but Proposition 5.3 uses  $\mathbb{F}_2$  coordinates and the delivered program uses  $\{0, 1\}^n$  with outputs  $\pm 1$  and the first  $m$  coordinates as the low bits. The source reuses letters as follows; no symbol was renamed.

| Symbol                               | Meaning here (where)                                                                                                                                                                                        | Not to be read as                                                                                                       |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| $neutral, R_f$                       | self-dual, (1); the change distance (2), so $R_f \geq 1$                                                                                                                                                    | anonymous (permutation invariant); a covering radius                                                                    |
| $\varepsilon_f(r), \varepsilon_n(r)$ | vulnerability $\mu_n(R_f \leq r)$ , (3)                                                                                                                                                                     | the tolerance $\epsilon > 0$ of the Chebyshev step in Section 9                                                         |
| $\eta, a, \delta_j$                  | the density tolerance of Theorem 6.1; $a = 1 - e^{-\eta}$ ; $\delta_j = a2^{1-j}$                                                                                                                           | $\eta$ of noisy-parity-cubes; the growth function $a(n)$ of Proposition 7.2; $a$ in $d = 2a + 1$ (proof of Theorem 2.1) |
| $d(n), d, d_i$                       | the largest odd integer $\leq n$ (also $d = 2a + 1$ ); the dimension of $\mathbb{F}_2^d$ in Proposition 5.3; $d_i = p_i - \frac{1}{2}$ in Section 9                                                         | Hamming distance, printed $d_H$                                                                                         |
| $m$                                  | a prefix length (Sections 4–6); $n = 2m + 1$ or $2m$ in Sections 2–3                                                                                                                                        | each other                                                                                                              |
| $C_r(f), C$                          | the robust positive core (6); the Hamming code of Proposition 5.3; an input cylinder (Theorems 5.1, 6.1, 7.1)                                                                                               | each other                                                                                                              |
| $D$                                  | the diameter bound of Theorem 2.3; $D = C \times C$ in Proposition 5.3                                                                                                                                      | each other                                                                                                              |
| $N, N_r(\cdot), \mathbb{N}$          | $N = 2^d - 1$ (Proposition 5.3) and an index in proofs; radius- $r$ neighbourhoods; the natural numbers                                                                                                     |                                                                                                                         |
| $A, A_n, A_+$                        | a fibre $F^{-1}(1)$ or a Borel set (Sections 8–9), the negative set in Lemma 4.2; $A_n = \{f_n = 1\}$ in (30) but $A_n = \sum_{i \leq n} b_i^2$ in Section 9; $A_+$ the event of eventually positive output | each other                                                                                                              |
| $b, b_i$                             | an output word (Theorem 6.1); $b_i = 2p_i - 1$ (Section 9)                                                                                                                                                  |                                                                                                                         |
| $F, F_\ell$                          | the infinite rule (46); the sharpness construction (22)                                                                                                                                                     | the finite label type $\mathbf{F}$ of the Lean file                                                                     |
| $H, h_\ell, h_L$                     | the decision map (33); majorities of tail votes                                                                                                                                                             | the family $\mathbb{H}$ of homeomorphisms in the Lean file                                                              |
| $E, E_0$                             | the divergence set (42); the finite-change equivalence relation                                                                                                                                             | each other                                                                                                              |
| $K(n, m, r), K$                      | the constant of Question Q1; an index in the proof of Theorem 7.1                                                                                                                                           |                                                                                                                         |
| $\tau_i, \kappa$                     | the flip of coordinate $i$ ; global complementation<br>$\kappa x = -x$                                                                                                                                      |                                                                                                                         |
| $\mu, \mu_n, \mu_p, \nu, \beta$      | fair product measure; uniform measure on $Q_n$ ; the biased product (48); $\nu = H_*\mu$ ; fair product on the output space                                                                                 |                                                                                                                         |

## 2 An exact solution for every finite committee

Let  $d(n)$  be the largest odd integer at most  $n$ , and use the convention that a sum with negative upper endpoint is zero.

**Theorem 2.1** (Exact finite robustness). *For every  $n \geq 1$  and integer  $r \geq 0$ ,*

$$\boxed{\max_{f \text{ neutral}} \rho_f(r) = 2^{1-d(n)} \sum_{j=0}^{(d(n)-1)/2-r} \binom{d(n)}{j}.} \quad (4)$$

*Majority on  $d(n)$  coordinates attains the maximum simultaneously for all  $r$ . In particular,*

$$\rho_{2m-1}^*(r) = \rho_{2m}^*(r), \quad (5)$$

*where the star denotes the optimum over neutral rules.*

The extra voter in an even committee therefore does not improve this extremal objective. The attaining rule may distinguish one voter and ignore that vote. Equation (5) is not a claim about every fairness criterion or every voting model.

## 2.1 The antipodal-core reduction

For a neutral rule, let

$$C_r(f) = \{x \in Q_n : f(x) = 1, R_f(x) > r\}. \quad (6)$$

The negative robust core is exactly  $-C_r(f)$ . Thus

$$\rho_f(r) = \frac{2|C_r(f)|}{2^n}. \quad (7)$$

**Lemma 2.2** (Small diameter of a robust core). *If  $x, y \in C_r(f)$ , then*

$$d_H(x, y) \leq n - 2r - 1. \quad (8)$$

*If  $2r \geq n$ , the core is empty.*

*Proof.* The Hamming balls  $\mathcal{B}_r(x)$  and  $\mathcal{B}_r(-y)$  carry opposite constant outputs, so they are disjoint. Two radius- $r$  balls in a Hamming cube intersect whenever the distance of their centres is at most  $2r$ : distribute the differing coordinates between the two centres to obtain a point within  $r$  of each. Consequently

$$2r < d_H(x, -y) = n - d_H(x, y),$$

which gives (8). If  $2r \geq n$ , even  $y = x$  would contradict this inequality.  $\square$

Here is the classical extremal input, stated in the exact form needed.

**Theorem 2.3** (Kleitman's diameter theorem [4, 5]). *If  $A \subseteq Q_n$  has diameter at most  $D < n$ , then*

$$|A| \leq \sum_{j=0}^s \binom{n}{j} \quad \text{if } D = 2s, \quad (9)$$

$$|A| \leq 2 \sum_{j=0}^s \binom{n-1}{j} \quad \text{if } D = 2s + 1. \quad (10)$$

*Both bounds are attained: respectively by a Hamming ball and by the product of a free coordinate with a Hamming ball in the remaining coordinates.*

[write] **5 October 2026.** Theorem 1.1 of [5] (arXiv version 1, read by the write) states exactly these two bounds for a set of diameter at most  $D < n$ , says that both are sharp, names the same extremal sets (for odd  $D$  also a Hamming ball of radius  $D/2$  centred at  $(\frac{1}{2}, 0, \dots, 0)$ ), and attributes the theorem to Kleitman. The bibliographic data of [4] were confirmed through its DOI; Kleitman's text was not read. (The bibliography of [5] gives a different reference for Kleitman's paper, *J. Combin. Theory Ser. A* **43** (1986), 85–90, which the write did not check.) The abstract of [10] writes the odd bound as  $\sum_{i \leq s} \binom{n}{i} + \binom{n-1}{s}$ , which equals (10) by Pascal's rule.

*Proof of Theorem 2.1.* First take  $n = 2m + 1$ . If  $r > m$ , Lemma 2.2 gives an empty core. Otherwise the core has diameter at most  $2(m - r)$ , so

$$|C_r(f)| \leq \sum_{j=0}^{m-r} \binom{n}{j}.$$

Equation (7) gives the upper bound.

For  $n = 2m$  and  $r \leq m - 1$ , the diameter is at most  $2(m - r - 1) + 1$ . The odd case of Kleitman's theorem gives

$$|C_r(f)| \leq 2 \sum_{j=0}^{m-r-1} \binom{n-1}{j},$$

and hence the same robust fraction as in dimension  $n - 1$ . Larger radii again give an empty core.

For attainment, let  $d = 2a + 1 = d(n)$  and define  $f(x) = \text{sgn}(x_1 + \dots + x_d)$ . If  $w$  of the first  $d$  coordinates are positive, then

$$R_f(x) = \frac{|2w - d| + 1}{2}.$$

The condition  $R_f > r$  is exactly  $w \leq a - r$  or  $w \geq a + r + 1$ . Counting these two symmetric tails proves (4). Coordinates ignored by the rule do not change its change distance. One and the same majority rule therefore attains all the bounds.  $\square$

## 2.2 Examples and the distinction from random-noise stability

For nine voters, the exact optimum is as follows. Ten voters give the same row.

| Budget $r$              | 0 | 1      | 2      | 3     | 4     |
|-------------------------|---|--------|--------|-------|-------|
| Optimal robust fraction | 1 | 65/128 | 23/128 | 5/128 | 1/256 |

For example, a nine-voter majority with five votes for one option is vulnerable to a single change, whereas one with six votes for that option survives every single change. The adversary is allowed to target the margin. This differs from the probability of agreement after independently resampling or flipping votes at random; no random-noise extremal theorem is being invoked.

The proof establishes an optimum and an attaining family. It does not classify all attaining Boolean rules. Even a complete classification of largest diameter cores must still be translated into a classification of the full decision rule, including its nonrobust region.

## 3 The scale of finite resilience

Take  $n = 2m + 1$  and write  $\rho_n^*$  for the optimal profile. If  $W_n \sim \text{Bin}(n, 1/2)$ , then

$$\rho_n^*(r) = 2\mathbb{P}(W_n \leq m - r). \quad (11)$$

The critical budget is of order  $\sqrt{n}$ , rather than of order  $n$ .

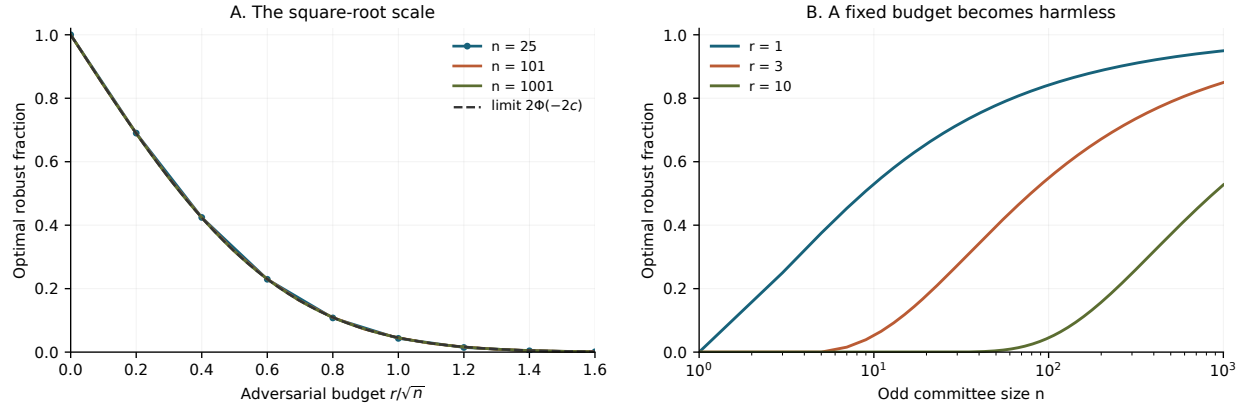


Figure 1: Exact finite optima and their scale. Left: the binomial formula at three odd committee sizes, plotted against  $r/\sqrt{n}$ , with its Gaussian limit. Right: resistance to fixed budgets improves as the committee grows. The figure is generated from exact binomial probabilities, not Monte Carlo estimates.

**Proposition 3.1** (Three regimes). *For integer budgets  $r_n \geq 0$ :*

(i) *If  $r_n = o(\sqrt{n})$ , then  $\rho_n^*(r_n) \rightarrow 1$ .*

(ii) *If  $r_n/\sqrt{n} \rightarrow c \in [0, \infty)$ , then*

$$\rho_n^*(r_n) \longrightarrow 2\Phi(-2c), \quad (12)$$

*where  $\Phi$  is the standard normal distribution function.*

(iii) *If  $r_n/\sqrt{n} \rightarrow \infty$ , then  $\rho_n^*(r_n) \rightarrow 0$ .*

*For integers  $1 \leq r = o(\sqrt{n})$ , more precisely,*

$$1 - \rho_n^*(r) = 2r\sqrt{\frac{2}{\pi n}} \left( 1 + O\left(\frac{r^2 + 1}{n}\right) \right). \quad (13)$$

*Proof.* The standardized variable  $2(W_n - n/2)/\sqrt{n}$  converges to a standard normal variable. Its cutoff in (11) is  $-(2r+1)/\sqrt{n}$ , which proves (12). For the finer estimate, the vulnerable weights are the  $2r$  consecutive integers  $m-r+1, \dots, m+r$ . Stirling's formula gives central mass  $\sqrt{2/(\pi n)}(1 + O(n^{-1}))$ . Ratios of adjacent binomial coefficients show that each of these  $2r$  masses differs relatively from a central mass by  $O((r^2 + 1)/n)$ , proving (13).

For completeness, a direct exponential bound proves the third regime without a normal approximation. For a sum  $S_n$  of  $n$  fair signs,  $\mathbb{E}e^{tS_n} = (\cosh t)^n \leq e^{nt^2/2}$ . Exponential Markov inequality, optimized at  $t = (2r+1)/n$ , gives

$$\rho_n^*(r) \leq 2 \exp\left(-\frac{(2r+1)^2}{2n}\right). \quad (14)$$

This tends to zero when the budget exceeds the square-root scale.  $\square$

An elementary bound is useful for explicit extraction schedules. Put  $c_m = \binom{2m}{m}/4^m$ . Induction from  $c_{m+1}/c_m = (2m+1)/(2m+2)$  gives  $c_m \leq 1/\sqrt{m+1}$ . The largest mass of  $\text{Bin}(2m+1, 1/2)$  is at most  $c_m$ . Therefore

$$\varepsilon_{\text{Maj}_n}(r) \leq \min\left\{1, 2r\sqrt{\frac{2}{n+1}}\right\}. \quad (15)$$

For  $r \leq m$  this counts  $2r$  central masses; for larger budgets the right-hand expression before truncation already exceeds one. In particular,  $n + 1 \geq 8r^2\delta^{-2}$  suffices for vulnerability at most  $\delta$ . The constant is deliberately elementary; (13) is sharper.

## 4 A sharp inequality for what early voters can predict

The key quantitative bridge does not require Kleitman's theorem. For  $1 \leq m \leq n$ , define

$$\alpha_f(m) = \|\mathbb{E}[f \mid \mathcal{F}_m]\|_\infty. \quad (16)$$

This is the largest possible conditional bias after any particular assignment of the first  $m$  votes. Also put

$$V(m, r) = \sum_{j=0}^{\min(m, r)} \binom{m}{j}. \quad (17)$$

**Theorem 4.1** (Sharp finite-prefix inequality). *For every neutral  $f : Q_n \rightarrow \{-1, 1\}$ , and every  $1 \leq r \leq m \leq n$ ,*

$$\alpha_f(m) \leq \min \left\{ 1, \frac{2^m}{V(m, r)} \varepsilon_f(r) \right\}. \quad (18)$$

*The coefficient  $2^m/V(m, r)$  is the best possible constant independent of the ambient dimension  $n$ , for each fixed pair  $(m, r)$ . In particular,*

$$\alpha_f(m) \leq \varepsilon_f(m), \quad \alpha_f(m) \leq \frac{2^m}{m+1} \varepsilon_f(1). \quad (19)$$

### 4.1 The smallest vulnerable region in a nonconstant cube

**Lemma 4.2.** *If  $h : Q_m \rightarrow \{-1, 1\}$  is nonconstant and  $1 \leq r \leq m$ , then*

$$|\{x : R_h(x) \leq r\}| \geq V(m, r). \quad (20)$$

*A rule with exactly one input of its minority color attains equality.*

*Proof.* The statement for  $m = 1$  is immediate, and for  $r \geq m$  every input is vulnerable. Proceed by induction, separating the last coordinate into two  $(m - 1)$ -dimensional slices.

If both slices are nonconstant, within-slice changes alone make at least  $2V(m - 1, r)$  vertices vulnerable. With the volume convention (17), this is at least  $V(m - 1, r) + V(m - 1, r - 1) = V(m, r)$ .

If one slice is constant, rename outputs so it is constantly positive. Let  $A$  be the nonempty negative set in the other slice. In that other slice, every point of its radius- $r$  neighborhood  $N_r(A)$  is vulnerable: positive points can reach  $A$ , and negative points can move to the positive constant slice with one change. In the constant slice, all points above  $N_{r-1}(A)$  are vulnerable. Since a neighborhood of a nonempty set contains a ball,

$$|N_r(A)| + |N_{r-1}(A)| \geq V(m - 1, r) + V(m - 1, r - 1) = V(m, r).$$

Oppositely colored constant slices make the entire cube vulnerable; identically colored constant slices are excluded by nonconstancy.

Finally, if the minority set is the singleton  $\{a\}$ , the vulnerable set is exactly  $\mathcal{B}_r(a)$  for  $r \geq 1$ . This proves attainment.  $\square$

## 4.2 Complementary fibres cancel

*Proof of the inequality in Theorem 4.1.* Write an input as  $(s, y)$ , where  $s \in Q_m$  is the prefix and  $y \in Q_{n-m}$  is the tail. Let  $B$  be the set of tails for which  $s \mapsto f(s, y)$  is nonconstant. On  $B^c$ , denote its common output by  $c(y)$ .

Neutrality implies that  $B$  is invariant under  $y \mapsto -y$ , and that  $c(-y) = -c(y)$  on  $B^c$ . Thus the integral of  $c$  over  $B^c$  vanishes. For each fixed prefix  $s$ ,

$$|\mathbb{E}_y f(s, y)| = |\mathbb{E}_y [\mathbf{1}_B(y) f(s, y)]| \leq \mathbb{P}_y(B). \quad (21)$$

When the tail has zero coordinates, the neutral full prefix fibre is nonconstant, so the same argument has no homogeneous part to cancel.

By Lemma 4.2, each bad tail contributes at least  $V(m, r)$  prefix inputs vulnerable to  $r$  prefix changes. Such changes are also legal changes in the whole input. Consequently

$$\varepsilon_f(r) \geq \mathbb{P}_y(B) \frac{V(m, r)}{2^m}.$$

Together with (21), this proves (18).  $\square$

## 4.3 Why the constant cannot be improved

Fix  $(m, r)$  with  $1 \leq r \leq m$ . Let  $h_\ell$  be majority on an odd number  $\ell$  of tail votes. Define

$$F_\ell(s, y) = \begin{cases} 1, & s = (1, \dots, 1), \\ -1, & s = (-1, \dots, -1), \\ h_\ell(y), & \text{otherwise.} \end{cases} \quad (22)$$

This is neutral and  $\alpha_{F_\ell}(m) = 1$ . For a fixed tail, the prefix rule is constant except for the corner forced to have the opposite output. Its  $r$ -vulnerable prefix region has exactly  $V(m, r)$  points.

If  $R_{h_\ell}(y) > r$ , no combination of at most  $r$  changes can alter the tail's majority. For such a tail, the  $r$ -vulnerable prefix inputs are exactly the radius- $r$  ball around the exceptional corner: they can reach that corner, or leave it if they start there. It follows that

$$\frac{V(m, r)}{2^m} \leq \varepsilon_{F_\ell}(r) \leq \frac{V(m, r)}{2^m} + \left(1 - \frac{V(m, r)}{2^m}\right) \varepsilon_{h_\ell}(r). \quad (23)$$

By Section 3, the last factor tends to zero. Hence  $\alpha_{F_\ell}(m)/\varepsilon_{F_\ell}(r) \rightarrow 2^m/V(m, r)$ . This proves sharpness uniformly over dimensions. It does not determine the best constant for every fixed triple  $(n, m, r)$ .

## 4.4 Conditional probabilities and comparisons between committees

For every positive-probability event  $A \in \mathcal{F}_m$ ,

$$\frac{1 - \alpha_f(m)}{2} \leq \mathbb{P}(f = 1 \mid A) \leq \frac{1 + \alpha_f(m)}{2}. \quad (24)$$

The estimate does not deteriorate when  $A$  is a very rare prefix event. For any earlier decision  $g : Q_m \rightarrow \{-1, 1\}$ ,

$$\left| \mathbb{P}(f = g) - \frac{1}{2} \right| \leq \frac{1}{2} \alpha_f(m) \leq \frac{2^{m-1}}{V(m, r)} \varepsilon_f(r). \quad (25)$$

Indeed,  $\mathbb{P}(f = g) = (1 + \mathbb{E}[fg])/2$ , and conditional expectation bounds the correlation. This is the precise reason strong resistance to changes in early votes can coexist with a later decision that is almost unrelated to the earlier one.

## 5 Why increasingly resilient decisions cannot settle

Let  $f_n$  be neutral and depend on the first  $n$  inputs, and put  $\varepsilon_n(r) = \varepsilon_{f_n}(r)$ . Distinguish two assumptions:

$$\varepsilon_n(1) \longrightarrow 0, \quad \text{one-vote resilience;} \quad (26)$$

$$\varepsilon_n(r) \longrightarrow 0 \quad \text{for every fixed } r, \quad \text{resilience at every fixed budget.} \quad (27)$$

The first is enough to prevent stabilization. The second allows us to arrange almost-sure divergence of robustness along an extracted subsequence.

**Theorem 5.1** (Mixing and unavoidable oscillation). *Under (26):*

(i) *For every  $g \in L^1(\mu)$ ,  $\mathbb{E}[f_n g] \rightarrow 0$ .*

(ii) *For every fixed measurable  $G : \Omega \rightarrow \{-1, 1\}$ ,*

$$\mathbb{P}(f_n = G) \longrightarrow \frac{1}{2}. \quad (28)$$

(iii) *The decisions take both signs infinitely often on a conull, comeager set. Their set of pointwise convergence is null and meager.*

(iv) *No deterministic infinite subsequence converges in probability.*

*Proof.* For each fixed  $m$ , Theorem 4.1 implies  $\alpha_{f_n}(m) \leq 2^m \varepsilon_n(1)/(m+1) \rightarrow 0$ . Given  $g \in L^1$ , conditional expectations on finite prefixes approximate  $g$  in  $L^1$ . More explicitly,

$$|\mathbb{E}[f_n g]| \leq \alpha_{f_n}(m) \|g\|_1 + \|g - \mathbb{E}[g \mid \mathcal{F}_m]\|_1. \quad (29)$$

First take  $n \rightarrow \infty$ , then  $m \rightarrow \infty$ . This proves (i), and the identity  $\mathbb{P}(f_n = G) = (1 + \mathbb{E}[f_n G])/2$  gives (ii).

Let  $A_+$  be the measurable event that  $f_n$  is eventually positive. On  $A_+$ ,  $f_n \rightarrow 1$ , so dominated convergence gives  $\mathbb{E}[\mathbf{1}_{A_+} f_n] \rightarrow \mu(A_+)$ . Part (i) makes this limit zero. Likewise the event of eventual negative output has measure zero.

For category, fix a nonempty input cylinder  $C$ , determined by the first  $m$  coordinates. For all sufficiently large  $n$ ,  $\alpha_{f_n}(m) < 1$ , so both signs occur on  $C$ . For each sign  $s$  and each  $N$ , the open set  $\bigcup_{n \geq N} \{f_n = s\}$  is therefore dense. Intersect these sets over  $s$  and  $N$ . This gives a dense  $G_\delta$  set on which both signs occur infinitely often.

All the arguments apply to each fixed deterministic infinite subsequence. If such a subsequence converged in probability, a further subsequence would converge almost surely to a measurable sign-valued limit  $G$ , contradicting (28). This proves (iv).  $\square$

Part (i) is weak-\* convergence to zero in  $L^\infty$ . For the sets  $A_n = \{f_n = 1\}$ , it says

$$\mu(A_n \cap B) \longrightarrow \frac{1}{2} \mu(B) \quad \text{for every fixed measurable } B. \quad (30)$$

This is mixing with density 1/2 in Rényi's terminology [6]. The zero weak limit is not an admissible decision, which must be a sign. The order of quantifiers matters:  $G$  in (28) is fixed before the limit. The statement is not uniform over a choice of  $G$  depending on  $n$ , and does not prohibit selecting a constant-sign subsequence after seeing the entire input.

### 5.1 An even weaker qualitative hypothesis

Let  $x^{(i)} = \tau_i x$  denote  $x$  with coordinate  $i$  flipped, and set  $\text{Inf}_i(f) = \mathbb{P}(f(X) \neq f(X^{(i)}))$ .

**Proposition 5.2.** *The conclusions of Theorem 5.1 remain true if*

$$\text{Inf}_i(f_n) \longrightarrow 0 \quad \text{for every fixed } i. \quad (31)$$

*Proof.* For a finite nonempty set  $S$ , put  $\chi_S(x) = \prod_{i \in S} x_i$ . For  $i \in S$ , invariance of product measure under a coordinate flip gives

$$\widehat{f}_n(S) = \mathbb{E}[f_n \chi_S] = \frac{1}{2} \mathbb{E}[(f_n - f_n \circ \tau_i) \chi_S],$$

so  $|\widehat{f}_n(S)| \leq \text{Inf}_i(f_n) \rightarrow 0$ . Neutrality gives  $\widehat{f}_n(\emptyset) = 0$ . The conditional expectation on a fixed  $m$ -prefix is the finite Walsh expansion over  $S \subseteq [m]$ , so its uniform norm tends to zero. The proof of Theorem 5.1 now applies.  $\square$

This statement says that if every particular early voter eventually loses influence, no fixed measurable eventual decision can emerge. It does not say that all small adversarial coalitions become harmless. For example,  $f_n(x) = x_n$  satisfies (31) but is always one-vote fragile.

### 5.2 One-vote resilience does not imply two-vote resilience

There is a sharper separation between (26) and (27). We give the construction because it rules out a potentially misleading inference in the puzzle.

**Proposition 5.3** (A sharp separation of budgets). *There are neutral rules of dimensions tending to infinity for which  $\varepsilon_f(1) \rightarrow 0$  but  $\varepsilon_f(2) = 1$  identically.*

*Proof.* Use binary coordinates temporarily. Let  $d \geq 2$ ,  $N = 2^d - 1$ , and index  $N$  coordinates by the nonzero vectors of  $\mathbb{F}_2^d$ . Define

$$C = \left\{ x \in \mathbb{F}_2^N : \sum_{v \neq 0} x_v v = 0 \right\}.$$

The syndrome map is surjective, so  $C$  has density  $1/(N+1)$ . A nonzero syndrome is corrected by flipping its indexed coordinate; hence every word lies within distance one of  $C$ . No nonzero codeword has weight one or two, because the columns are nonzero and distinct. Thus the minimum distance is at least three. Finally  $\sum_{v \neq 0} v = 0$ , so  $C$  is invariant under complementation.

The product  $D = C \times C$  has density  $(N+1)^{-2}$ , minimum distance at least three, covering radius at most two, and is complement-invariant. Its radius-one balls are disjoint, giving

$$u_N := \mathbb{P}((X, Y) \in N_1(D)) = \frac{2N+1}{(N+1)^2}.$$

Let  $h_L$  be majority of an odd number  $L$  of further fair signs, and set

$$f(X, Y, Z) = h_L(Z)(-1)^{\mathbf{1}_D(X, Y)}.$$

Complementation leaves  $D$  invariant and reverses  $h_L$ , so  $f$  is neutral. Every point outside  $D$  can enter it with at most two changes; every point of  $D$  can leave with one. Thus  $R_f \leq 2$  everywhere.

One-change vulnerability occurs exactly when the code coordinates lie in  $N_1(D)$ , or the majority is one-change vulnerable. These events use disjoint input coordinates. Writing  $t_L = \varepsilon_{h_L}(1)$ , we obtain

$$\varepsilon_f(1) = u_N + t_L - u_N t_L \longrightarrow 0 \quad (32)$$

as  $d, L \rightarrow \infty$ . Taking, for example,  $L = N$  gives the claimed sequence of dimensions. Padding by ignored coordinates supplies a rule at every intermediate horizon if desired.  $\square$

## 6 Extracting a continuous open stream of almost fair decisions

Oscillation alone leaves substantial freedom. We can arrange much more: a fixed subsequence of horizons has an output law uniformly comparable with a product of fair coins, while every output stream remains topologically attainable.

**Theorem 6.1** (Quantitative open recoding). *Suppose  $f_n$  is neutral and  $\varepsilon_n(1) \rightarrow 0$ . For every  $\eta > 0$  there is a deterministic strictly increasing sequence  $n_1 < n_2 < \dots$  such that the map*

$$H : \Omega \longrightarrow \Omega, \quad H(x) = (f_{n_j}(x))_{j \geq 1} \quad (33)$$

*is continuous, open, and onto. If  $\nu = H_*\mu$  and  $\beta$  is fair product measure on the output space, then*

$$e^{-\eta}\beta(A) \leq \nu(A) \leq e^{\eta}\beta(A) \quad \text{for every Borel } A. \quad (34)$$

*Equivalently,*

$$e^{-\eta} \leq \frac{d\nu}{d\beta} \leq e^{\eta} \quad \beta\text{-almost everywhere.} \quad (35)$$

*If the stronger assumption (27) holds, the same selection can also ensure*

$$R_{f_{n_j}}(x) \longrightarrow \infty \quad \text{for } \mu\text{-almost every } x. \quad (36)$$

### 6.1 The deterministic selection and the likelihood bounds

*Proof.* Let  $a = 1 - e^{-\eta}$ , and for  $j \geq 2$  put  $\delta_j = a2^{1-j}$ . Thus  $0 < \delta_j < 1$  and  $\sum_{j \geq 2} \delta_j = a$ . Choose any first horizon. Once  $n_{j-1}$  has been chosen, take  $n_j > n_{j-1}$  so large that

$$\alpha_{f_{n_j}}(n_{j-1}) \leq \delta_j. \quad (37)$$

This is possible by Theorem 4.1. Under (27), additionally require

$$\varepsilon_{n_j}(\max\{n_{j-1}, j\}) \leq \delta_j; \quad (38)$$

this stronger condition already implies (37).

The first selected output is exactly fair by neutrality. Any history of preceding selected outputs is an event in  $\mathcal{F}_{n_{j-1}}$ . By (24), every next conditional probability lies between  $(1 - \delta_j)/2$  and  $(1 + \delta_j)/2$ . For every finite output word  $b = (b_1, \dots, b_J)$ ,

$$\prod_{j=2}^J (1 - \delta_j) \leq \frac{\nu([b])}{2^{-J}} \leq \prod_{j=2}^J (1 + \delta_j). \quad (39)$$

The elementary inequalities

$$\prod (1 - \delta_j) \geq 1 - \sum \delta_j = e^{-\eta}, \quad \prod (1 + \delta_j) \leq e^{\sum \delta_j} \leq e^{\eta}$$

give uniform bounds. They extend from cylinder sets and their finite unions to all Borel sets, for example by regularity of finite measures on Cantor space. This proves (34), and the Radon–Nikodym theorem gives (35).

Under (38),  $\sum_j \mathbb{P}(R_{f_{n_j}} \leq j) < \infty$ . Borel–Cantelli implies that  $R_{f_{n_j}} > j$  eventually almost surely, proving (36).  $\square$

The density estimate is stronger than just convergence of finite-dimensional distributions. It compares probabilities of *all* Borel properties of the whole output stream. In particular  $\|\nu - \beta\|_{\text{TV}} \leq e^{\eta} - 1$ . Nevertheless, the selected outputs need not be independent; equivalence with product measure is the claim.

## 6.2 Why the decision map is open and onto

*Completion of the topological part of Theorem 6.1.* Every output coordinate depends on finitely many input coordinates, so  $H$  is continuous. Let  $C$  be an input cylinder fixing the first  $k$  votes, and choose  $J$  with  $n_J \geq k$ . Fix an output word  $b$  of length  $J$  compatible with  $C$ . Then

$$A = C \cap \{f_{n_1} = b_1, \dots, f_{n_J} = b_J\}$$

is a nonempty event in  $\mathcal{F}_{n_J}$ , and hence has positive measure. The conditional bounds, which are strictly between zero and one, permit either next sign on  $A$ . Repeating the argument permits every finite continuation. The resulting nonempty closed constraints are nested in the compact input space, so every infinite continuation is realized.

It follows that  $H(C)$  is exactly the union of the output cylinders  $[b]$  over all length- $J$  words compatible with  $C$ . This is a finite union of cylinders, and is open. Hence  $H$  is an open map. Equation (39) gives a preimage for every finite output word; compactness again realizes every infinite output stream. Thus  $H$  is onto.  $\square$

## 6.3 A coupling interpretation

On an enlargement of the probability space, there are iid fair signs  $Z_j$  such that, writing  $Y_j = f_{n_j}(X)$ ,

$$\sum_{j \geq 2} \mathbb{P}(Y_j \neq Z_j) \leq \frac{1}{2} \sum_{j \geq 2} \delta_j < \infty. \quad (40)$$

One way to see this is to construct the pair sequentially. Given the already constructed pair histories, let the next  $Y_j$  have its true conditional Bernoulli law, which depends on the previous  $Y$ 's, and couple it maximally with a fair  $Z_j$ . The conditional mismatch probability is at most  $\delta_j/2$ , while the conditional marginal of  $Z_j$  is always fair. This produces the prescribed law of the  $Y$ -stream and an iid  $Z$ -stream. Standard disintegration allows the original input to be retained as well. Borel–Cantelli then makes the two streams differ at only finitely many positions almost surely.

The enlarged probability space is part of this coupling statement. It does not assert that an additional independent sequence was already available as a deterministic function of the input. The bounded-density and openness conclusions above are direct properties of the deterministic map  $H$ .

## 6.4 An explicit schedule for ordinary majorities

For  $f_n = \text{Maj}_n$  at odd horizons, set  $q_j = \max\{n_{j-1}, j\}$ . By (15), any odd  $n_j > n_{j-1}$  satisfying

$$n_j + 1 \geq 8q_j^2 \delta_j^{-2} \quad (41)$$

meets (38). This produces a completely specified, although deliberately sparse, subsequence. Finding much denser sequences with comparably strong infinite-stream guarantees is a meaningful further problem.

## 7 Opposite answers from probability and category

Assume (27), and use the stronger selection in Theorem 6.1. Define

$$E = \{x : R_{f_{n_j}}(x) \rightarrow \infty\}. \quad (42)$$

**Theorem 7.1** (Two notions of typical input). *The selected horizons have the following properties.*

(i) For almost every input,  $R_{f_{n_j}} \rightarrow \infty$ , and the output stream is normal: every finite sign word of length  $k$  has limiting overlapping frequency  $2^{-k}$ .

(ii) On a comeager set of inputs,

$$\liminf_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J \mathbf{1}_{\{f_{n_j}=1\}} = 0, \quad \limsup_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J \mathbf{1}_{\{f_{n_j}=1\}} = 1, \quad (43)$$

and  $R_{f_{n_j}} = 1$  infinitely often.

(iii) The set  $E$  is Borel, conull, meager, and invariant under finite changes and global complementation. For  $x, y \in E$ ,

$$x E_0 y \implies H(x) E_0 H(y). \quad (44)$$

*Proof.* The radius conclusion in (i) is (36). Normality has full fair-product measure. To verify this standard fact directly, fix a word of length  $k$ . Its occurrence indicators at positions in any fixed residue class modulo  $k$  depend on disjoint input blocks and are iid with mean  $2^{-k}$ . Apply the strong law to each of the  $k$  classes, then to the countably many words. The measure equivalence (35) transfers this full-measure property to  $H(X)$ .

For (ii), in the output Cantor space the set of sequences with frequencies having liminf zero and limsup one is a dense  $G_\delta$ . For example, for each  $q, N \geq 1$ , the requirement that some prefix of length at least  $N$  has positive frequency below  $1/q$  is open and dense: append a sufficiently long block of negative signs. The analogous upper requirement is also open and dense. Intersect them. A continuous open surjection pulls dense open sets back to dense open sets, so (43) holds on a comeager input set.

For recurring fragility, fix a nonempty input cylinder  $C$ . At all sufficiently large selected horizons, both decisions occur on  $C$ , since its conditional bias is below one. The completions of  $C$  to that horizon form a connected finite cube. A path between opposite decisions contains a bichromatic edge, and its endpoints have  $R_f = 1$ . Hence, for every  $K$ ,

$$\bigcup_{j \geq K} \{x : R_{f_{n_j}}(x) = 1\}$$

is open and dense. Intersect over  $K$ . This also shows that  $E$  is meager. Its Borel character follows directly from (42).

Finally, suppose  $x$  and  $y$  differ in at most  $s$  coordinates. Whenever  $R_f(x) > s$ , the outputs agree and

$$R_f(y) \geq R_f(x) - s. \quad (45)$$

Indeed, an opposite-output point within distance less than  $R_f(x) - s$  of  $y$  would contradict the definition of  $R_f(x)$ , by the triangle inequality. Thus  $x \in E$  implies  $y \in E$ , and the two output streams eventually agree. Neutrality preserves all change distances under complementation. This proves (iii).  $\square$

[write] **A step made explicit, 5 October 2026.** The recurring-fragility step uses, for a cylinder  $C$  fixing the first  $k$  votes, that  $\alpha_{f_{n_j}}(k) < 1$  once  $n_{j-1} \geq k$ . This follows from (37) because  $\alpha_f$  is nondecreasing: for  $k \leq m$  the tower property gives  $\mathbb{E}[f \mid \mathcal{F}_k] = \mathbb{E}[\mathbb{E}[f \mid \mathcal{F}_m] \mid \mathcal{F}_k]$ , and conditional expectation does not increase the uniform norm, so  $\alpha_f(k) \leq \alpha_f(m)$ . Hence  $\alpha_{f_{n_j}}(k) \leq \alpha_{f_{n_j}}(n_{j-1}) \leq \delta_j < 1$ . Since  $C$  is an atom of  $\mathcal{F}_k$ , the average of  $f_{n_j}$  over  $C$  has absolute value at most  $\alpha_{f_{n_j}}(k) < 1$ , so  $f_{n_j}$  takes both values on  $C$ .

| Property                      | Almost every input                       | A comeager set of inputs                         |
|-------------------------------|------------------------------------------|--------------------------------------------------|
| Selected decision frequencies | Every finite word has its fair frequency | Positive frequency has $\liminf 0$ , $\limsup 1$ |
| Selected change distances     | Tend to infinity                         | Equal 1 infinitely often                         |
| Eventual decision             | Does not exist                           | Does not exist                                   |

There is no contradiction: a conull set may be meager. Also, (44) is a homomorphism of equivalence relations on  $E$ , not a reduction; no converse is asserted. The restriction  $H|_E$  is not claimed to remain open or onto.

### 7.1 Why no fair-frequency conclusion holds for all horizons

**Proposition 7.2** (Arbitrarily slow changes and wild full-sequence frequencies). *There are neutral rules satisfying (27) for which the positive-decision frequency along all horizons has  $\liminf$  zero and  $\limsup$  one almost surely. Moreover, for any prescribed nondecreasing unbounded function  $a : \mathbb{N} \rightarrow \mathbb{N}$ , they can be chosen so that, for every input, the number of decision changes up to horizon  $n$  is at most  $a(n)$  for all sufficiently large  $n$ .*

*Proof.* Partition successive input coordinates into disjoint odd blocks of lengths  $\ell_j \rightarrow \infty$ , and let  $Z_j$  be their majorities. The  $Z_j$ 's are independent fair signs. Choose deterministic release horizons  $t_j$  so large that the first  $j$  blocks have been read by  $t_j$ ,  $a(t_j) \geq j$ , and  $t_{j+1}/t_j \rightarrow \infty$ . Define  $f_n = Z_j$  for  $t_j \leq n < t_{j+1}$ , and use  $f_n(x) = x_1$  before  $t_1$ .

At each horizon the rule only uses available votes. Its robustness profile is that of  $\text{Maj}_{\ell_j}$ , so (27) holds. Changes can occur only at the release horizons, with no changes before  $t_1$ . There are at most  $j$  changes during stage  $j$ , which is at most  $a(n)$  for  $n \geq t_j$ . At horizon  $t_{j+1} - 1$ , the last constant block of decisions occupies a fraction tending to one of the entire elapsed time. Since the independent signs  $Z_j$  take both values infinitely often almost surely, the full-sequence frequency approaches both zero and one along subsequences.  $\square$

Thus infinite oscillation need not have a universal rate. A rate question must also impose quantitative resilience, restrictions on ignored votes, or regular growth conditions on the rules.

## 8 A perfect infinite rule exists under choice

The direct infinite version asks for  $F : \Omega \rightarrow \{-1, 1\}$  satisfying

$$x E_0 y \implies F(x) = F(y), \quad F(-x) = -F(x). \quad (46)$$

The first condition says that every finite intervention is harmless, at every input. This is an exact, global requirement, stronger than the probabilistic finite-horizon hypotheses.

**Theorem 8.1** (Existence and complete failure of regularity). *In ZFC there is an everywhere-defined rule satisfying (46). No such rule is measurable for completed fair-product measure, and neither fibre has the Baire property. In fact both fibres have inner measure zero and outer measure one.*

*Existence.* Consider the quotient set  $\Omega/E_0$ . Complementation induces a fixed-point-free involution on it: an infinite stream and its complement disagree at every coordinate. Choose one class from each pair of complementary classes. Give all streams in the chosen classes output 1, and their complementary classes output  $-1$ . This defines the required rule.  $\square$

This proof uses choice for a particular family of pairs; it does not identify the assertion with full choice. Another sufficient principle is the existence of a nonprincipal ultrafilter  $\mathcal{U}$  on  $\mathbb{N}$ . Then

$$F(x) = 1 \iff \{i : x_i = 1\} \in \mathcal{U} \quad (47)$$

is neutral and finite-change invariant. It is also monotone in the votes. This gives a concrete set-theoretic form of an infinite decision, while leaving its exact weak-choice strength as a separate issue.

*Measure obstruction.* First prove the relevant zero–one law. If a measurable set  $A$  is invariant under finite coordinate flips, all cylinders of the same finite length have the same conditional probability of  $A$ . Thus  $A$  is independent of every finite-prefix event, and by approximation of measurable sets by cylinder events, it is independent of itself. Hence  $\mu(A) = \mu(A)^2$ .

If  $A = F^{-1}(1)$  were measurable, this zero–one law would give  $\mu(A) \in \{0, 1\}$ . Complementation preserves  $\mu$  and takes  $A$  to  $A^c$ , so neutrality instead gives  $\mu(A) = 1/2$ .

For the inner-measure claim, suppose a measurable  $B \subseteq A$  had positive measure. Saturate  $B$  under the countable group of finite flips. The saturation is measurable, invariant, still contained in  $A$ , and has measure one by the zero–one law. Its complement image is a disjoint measure-one subset of  $A^c$ , impossible. Thus  $\mu_*(A) = 0$ . The same holds for  $A^c$ , which implies  $\mu^*(A) = \mu^*(A^c) = 1$ .  $\square$

*Category obstruction.* An invariant set with the Baire property is either meager or comeager. Indeed, if it is nonmeager, it is comeager inside some basic cylinder. Finite flips take that cylinder to the finitely many cylinders of the same length covering the whole space. Invariance makes the set comeager in each, hence comeager globally. Neutrality interchanges the two fibres by a homeomorphism, so they cannot be one meager and one comeager. They also cannot both be meager or both be comeager in Cantor space. Thus neither has the Baire property.  $\square$

The existence of  $F$  does not contradict Theorem 5.1. An arbitrary choice function can have no measurable pointwise approximation on a positive-measure set of the specified kind. Finite combinatorial solvability and set-theoretic existence do not by themselves provide regular limits.

## 9 How a small statistical bias changes the infinite answer

The perfect rule remains a global object satisfying (46), but change the law of the votes. Let  $0 < p_i < 1$  and

$$\mu_p = \bigotimes_{i \geq 1} \text{Bernoulli}(p_i), \quad \bar{p}_i = 1 - p_i, \quad (48)$$

where each Bernoulli law is on  $\{-1, 1\}$ , assigning probability  $p_i$  to 1 and probability  $1 - p_i$  to  $-1$ . Neutrality still concerns renaming the alternatives; the evaluation law is now permitted to favor one.

**Theorem 9.1** (Exact bias threshold). *In ZFC, there is an everywhere-defined rule satisfying (46) that is measurable for the completion of  $\mu_p$  if and only if*

$$\boxed{\sum_{i=1}^{\infty} (p_i - \tfrac{1}{2})^2 = \infty.} \quad (49)$$

*Even in the affirmative case, no such globally defined rule is Borel or has the Baire property.*

### 9.1 The product-measure input

Kakutani's theorem [9] says that these two product measures are either mutually absolutely continuous or mutually singular, and are equivalent exactly when

$$\prod_{i=1}^{\infty} 2\sqrt{p_i(1-p_i)} > 0. \quad (50)$$

Writing  $d_i = p_i - 1/2$ , we have

$$2d_i^2 \leq 1 - \sqrt{1 - 4d_i^2} \leq 4d_i^2.$$

For a sequence of numbers in  $(0, 1]$ , its product is positive exactly when the sum of its deficits from one is finite. Thus

$$\mu_p \sim \mu_{\bar{p}} \iff \sum_i d_i^2 < \infty; \quad \mu_p \perp \mu_{\bar{p}} \iff \sum_i d_i^2 = \infty. \quad (51)$$

*Proof of Theorem 9.1.* Suppose first that the measures are equivalent. A completed-measurable finite-change-invariant set obeys the product zero-one law, so if  $A = F^{-1}(1)$ , then  $\mu_p(A)$  is zero or one. Complementation maps  $\mu_p$  to  $\mu_{\bar{p}}$  and maps  $A$  to  $A^c$ . Hence  $\mu_{\bar{p}}(A) = 1 - \mu_p(A)$ , contradicting equivalence, which preserves null and conull sets. The zero-one law here follows by deleting any finite prefix. All sections of an exactly invariant set coincide. Since every prefix has positive probability, completed measurability gives completed-measurable tail sections. The set is therefore independent of every finite-prefix event; approximation implies independence of itself. This proves the zero-one law after completion as well.

Now suppose the measures are singular. Take a Borel set  $A_0$  with  $\mu_p(A_0) = 1$  and  $\mu_{\bar{p}}(A_0) = 0$ . Every finite flip is nonsingular for both measures, since all coordinate probabilities are strictly between zero and one. Intersect all finite-flip translates of  $A_0$  to obtain a Borel, exactly finite-change-invariant set  $A$  with the same two measure values. Let  $\kappa x = -x$  and put

$$A' = A \setminus \kappa A.$$

Then  $A'$  and  $\kappa A'$  are disjoint invariant Borel sets,  $\mu_p(A') = 1$ , and  $\mu_p(\kappa A') = 0$ . Assign opposite decisions on these two sets. Their complement is invariant under finite changes and under  $\kappa$ , and is null for both product measures. On its remaining complementary  $E_0$ -class pairs, use the choice construction of Theorem 8.1.

The resulting full-domain rule is measurable in the completion of  $\mu_p$ , because the arbitrary part lies in a Borel null set. The equivalence (51) gives (49). Global Borel or Baire regularity is still impossible by Theorem 8.1, independently of which law is used to evaluate a particular completion.  $\square$

For example, with  $p_i = 1/2 + c(i+1)^{-\gamma}$ ,  $0 < c < 1/2$  and  $\gamma \geq 0$ , the answer is affirmative exactly when  $\gamma \leq 1/2$ . An arbitrarily small bias can therefore matter if its squared deviations accumulate without bound. This extension is a deduction from a classical product-measure theorem, not a new version of Kakutani's theorem. It concerns existence in one completed biased law; it does not produce a universally measurable neutral selector.

### 9.2 An explicit measurable part of the biased rule

The conull part in the affirmative case need not be obtained by an abstract singular-measure separator. Put  $b_i = 2p_i - 1$ ,

$$A_n = \sum_{i \leq n} b_i^2, \quad S_n(x) = \sum_{i \leq n} b_i x_i,$$

and choose  $n_j$  as the first index with  $A_{n_j} \geq j^2$ . Under  $\mu_p$ , the mean of  $S_{n_j}$  is  $A_{n_j}$ , and its variance is at most  $A_{n_j}$ . Hence, for every  $\epsilon > 0$ ,

$$\mu_p \left( \left| \frac{S_{n_j}}{A_{n_j}} - 1 \right| > \epsilon \right) \leq \frac{1}{\epsilon^2 j^2}.$$

Chebyshev's inequality and Borel–Cantelli, for rational positive  $\epsilon$ , show that  $S_{n_j}/A_{n_j} \rightarrow 1$  almost surely. Under the complemented law, the limit is  $-1$  almost surely. The two corresponding Borel limit sets are disjoint and finite-change invariant: a finite edit changes each sufficiently late numerator by a fixed finite amount, while the denominator diverges. They are exchanged by complementation. Assign opposite decisions on these two sets and use choice only on the remaining null set. This makes the statistical part of the construction explicit.

## 10 Computation, reproducibility, and the ProveIt connection

### 10.1 What the supplied computation checks

The accompanying `code/verify.py` uses integer arithmetic and the Python standard library. It enumerates every neutral Boolean rule in dimensions one through five. A neutral rule has one free output choice on each antipodal pair, so the numbers of rules are

$$2, \quad 4, \quad 16, \quad 256, \quad 65\,536,$$

for a total of 65 814. The program computes Hamming erosions of each positive fibre, obtaining the complete exact robustness profile. It verifies the envelope (4) and the prefix inequalities with exact integer comparisons. Its larger-majority tables also use exact rational values.

The successful exhaustive run includes 394 576 checks of  $\alpha_f(m) \leq \varepsilon_f(m)$ , and 985 710 checks of the general volume inequality (18). These counts overlap at  $r = m$ . An independent direct minimum-Hamming-distance calculation agrees with the erosion method for all 278 neutral maps in dimensions one through four.

An erosion step replaces a set  $A$  by the points whose own vertex and all neighbors lie in  $A$ . Iterating it  $r$  times gives exactly the points whose entire Hamming radius- $r$  ball lies in  $A$ . This supplies a second concrete view of the robust core, and an executable finite certificate.

These calculations check the finite statements and expose normalization or off-by-one errors; they are not a replacement for the proofs in arbitrary dimension. The probability and category theorems are proved mathematically, not inferred from simulations. The package contains the verifier, its actual JSON report, and the exact numerical table used for examples.

### 10.2 The source snapshot and what is already formalized

The repository inspection used the fixed ProveIt revision

$$3b9458b31f459b480ab43a068d7cc56f4cfdbb28.$$

This avoids ambiguity from the actively changing main branch. The relevant source path is

$$\text{SetTheory/Cardinals/Cardinals/Countable/GenericErgodicity.lean.}$$

The inspected declarations include:

| Declaration                            | Role in this article                                                             |
|----------------------------------------|----------------------------------------------------------------------------------|
| <code>zero_one</code>                  | Topologically transitive invariance forces a Baire set to be meager or comeager. |
| <code>generic_const</code>             | A regular invariant map to a finite set is generically constant.                 |
| <code>fixed_label</code>               | A further symmetry must fix that generic output.                                 |
| <code>Cantor.transitive</code>         | Finite coordinate changes act topologically transitively on binary streams.      |
| <code>Cantor.global_fixed_point</code> | An existing specialization to index permutations and finite label actions.       |

The last specialization uses *index permutations*; global bit complementation is a different homeomorphism. The complement-selector obstruction in this article follows from the abstract declarations, with that homeomorphism supplied separately. It should not be reported as an already named, directly matching repository theorem. The source presents proofs without admitted statements; this investigation read the source but did not rebuild it.

The neighboring finite-symmetry file `SetTheory/Cardinals/Cardinals/Combinatorics/FiniteCycles.lean` provides additional conceptual motivation through equivariant-selector obstructions. It is not the source of the Hamming extremum. Targeted repository searches did not locate the quantitative robustness-to-prefix-predictability development proved here. That is evidence about the inspected material, not a certification of novelty across all literature or repository history.

[write] **Checked against the repository, 5 October 2026.** At the commit this write is based on, `GenericErgodicity.lean` is byte-identical to its state at the pin (blob c9b5ce93), and so is `FiniteCycles.lean` (blob b1da77ef). The five declarations of the table exist with the roles stated, in the namespace `Cardinals.Countable` (the last two in its sub-namespace `Cantor`); the file’s header says “Everything is proved; nothing is admitted”; the library root `SetTheory/Cardinals/Cardinals.lean` imports it; and the library’s README lists these declarations as the proved formalization of Proposition 15.3 of the large-cardinal synthesis. No Lean build was run for this report. `Cantor.global_fixed_point` does act through permutations of the index set, and global complementation is not among the homeomorphisms the file defines. The intake’s searches found no repository file other than this report that treats Kleitman’s diameter theorem or the worst-case (adversarial) Hamming robustness of neutral decision rules (the only other occurrence of Kleitman’s name in a report, in bounded-width-power-set-compression, cites the Greene–Kleitman symmetric-chain decomposition), so the statement above is still true.

*Remark 10.1* ([write] The category half from the Lean declarations, 5 October 2026). Identify  $\Omega$  with the Cantor space  $\mathbb{N} \rightarrow \text{Bool}$  of the Lean file, a compact Hausdorff and hence Baire space, and let  $F$  satisfy (46). Suppose one fibre of  $F$  has the Baire property. The other fibre is its complement, so both fibres are Baire measurable. (1) The finite flips `Cantor.flip S` ( $S$  a finite set of coordinates) are homeomorphisms acting topologically transitively (`Cantor.transitive`), and  $F$  is invariant under each of them by the first condition of (46); `generic_const` therefore gives a label  $\ell_0$  with  $F = \ell_0$  on a comeager set. (2) Complementation  $\kappa$  is a homeomorphism and  $F \circ \kappa = -F$  everywhere; `fixed_label`, applied with  $\sigma = \kappa$  and  $\rho$  the negation of labels, gives  $-\ell_0 = \ell_0$ , which is impossible. So neither fibre has the Baire property: this is the category half of Theorem 8.1, by the route the source describes above. *Formal status.* The three declarations are proved in Lean; complementation as a Lean homeomorphism and the assembled statement are not in the repository, so, as the source says, this is not a named repository theorem. The measure half of Theorem 8.1 and every finite statement of this report have no Lean counterpart.

### 10.3 A practical formalization order

The following order separates finite algebraic obligations from measure and category arguments.

1. Define neutrality, finite Hamming balls, change distance, and erosions. Prove that robust negative cores are antipodal images of robust positive cores.
2. Formalize Lemma 2.2 and the majority count. State the exact external dependency on Kleitman’s diameter theorem until that theorem itself is formalized; do not hide it behind an unrestricted axiom.
3. Prove Lemma 4.2 by dimension induction. This gives a fully elementary finite component and is well suited to independent Lean and Rocq developments.
4. Prove the fibre-cancellation identity and Theorem 4.1 using finite uniform sums. Its sharpness uses only the majority estimate.
5. Transfer finite rules to product space. Establish the conditional expectation bounds, the  $L^1$  approximation lemma, and mixing.
6. Separate the null and meager convergence proofs. Reuse the repository’s generic-ergodicity framework for the direct infinite-rule obstruction.
7. Build the deterministic extraction, cylinder likelihood estimates, and compactness proof of openness. Formalize the finite-edit-invariant conull domain as an additional statement, not as a claim about all inputs.

If a family of finite truth tables is computable, the extraction indices can be found by exhaustive search for the required rational inequalities, under the proved promise that they eventually hold. This is an effective search procedure, but no efficient complexity bound follows without a quantitative modulus of resilience. For majorities, (41) supplies an explicit modulus directly.

## 11 Further research questions and topics

The questions below extend the designed problem. A question labeled a *target* is not being asserted to be a previously published open problem; its wider literature status requires a focused audit. The cited stability and subsequence literature should be consulted before priority claims.

**Q1. Fixed-dimension sharp prefix constants.** Determine the exact function

$$K(n, m, r) = \sup_{f \text{ neutral}} \frac{\alpha_f(m)}{\varepsilon_f(r)}.$$

Theorem 4.1 determines  $\sup_{n \geq m} K(n, m, r)$ , but generally uses increasing ambient dimension for sharpness. The exact finite-dimensional correction is a concrete extremal target amenable to certificates.

**Q2. Classification and stability of optimal decision rules.** For fixed  $(n, r)$ , classify all neutral maximizers of  $\rho_f(r)$ , then relate near-maximal robustness to distance from a majority rule after cube symmetries. Existing diameter stability results, including [10], concern the extremal cores; a full-rule classification must also control the intervening boundary region. A statement that only repeats known anticode stability would not constitute progress.

- Q3. The joint tradeoff between two intervention budgets.** Determine attainable pairs  $(\varepsilon_f(1), \varepsilon_f(2))$ , and more generally entire profiles, subject to neutrality and dimension. Proposition 5.3 shows that small one-vote vulnerability alone gives no asymptotic two-vote guarantee. Quantitative restrictions on rule complexity, monotonicity, or anonymity may restore implications.
- Q4. Dense extraction from majority.** Find the slowest growth of  $n_j$  that allows the majority output law to be equivalent to Bernoulli measure, or to have uniformly bounded density as in (35). The sufficient schedule (41) is not proposed as optimal. Distinguish equivalence of full-stream laws from pairwise decorrelation and from a central limit theorem.
- Q5. A quantitative cost of changing one’s mind.** Impose a specified estimate such as  $\varepsilon_n(r) \leq Cr/\sqrt{n}$  over a stated range, and bound the smallest possible number of decision changes up to horizon  $n$ . Proposition 7.2 rules out a rate theorem based only on qualitative resilience. Restrictions requiring substantial use of all available voters may be needed.
- Q6. Several alternatives and other symmetry groups.** Replace the two labels by a transitive finite group action, and formulate neutrality so that no fixed label is allowed. The infinite generic-ergodicity obstruction has a natural generalization, but the finite robust cores form a multi-part packing problem. Determine which groups and alphabets admit a sharp volume formula and a counterpart of the prefix inequality.
- Q7. Other adversarial metrics.** Study weighted vote changes, interval edits, deletion/insertion distance, or graph-local interventions. The proof strategy suggests two separate objects: the minimum vulnerable fraction in a nonconstant fibre, and the largest diameter or separated-core family. Identifying these two invariants may reveal which finite–infinite conclusions survive.
- Q8. The exact weak-choice principle.** Locate the existence of an everywhere neutral  $E_0$ -invariant two-label rule among weak choice principles. Choice for the relevant quotient pairs and a nonprincipal ultrafilter are sufficient by different constructions; no equivalence with either is established here. Monotonicity or finite-intersection preservation changes the question and must be specified explicitly.
- Q9. Beyond independent biased inputs.** For dependent laws  $\mu$ , investigate whether equivalence or singularity of  $\mu$  and its complemented law still characterizes completed-measure selectors. Theorem 9.1 uses both product tail triviality and nonsingularity of finite flips. Counterexamples should isolate which of those hypotheses fails.
- Q10. Effective and formal quantitative ergodicity.** Formalize the prefix inequality, code separation, and open recoding in ProveIt. Develop a reusable interface turning finite vulnerability bounds into measurable and Baire nonconvergence theorems, with explicit assumptions about measure completion and choice.

### 11.1 Further questions and research

[write] **Added 5 October 2026, batch 114.** Following Vladimir Reshetnikov’s standing rule of 4 October 2026, every claim of the source that is not proved in full is stated here as an open question, with its source, the argument offered and what is missing. Questions Q1–Q10 above are the source’s own research proposals, and all ten are open. Four sentences of the body that point beyond what is proved are attached to the question they belong to: the end of Section 2 (classification, Question 2), the end of the explicit schedule in Section 6 (denser sequences, Question 4), the paragraph after Proposition 7.2 (rates, Question 5) and the “natural generalization” of the infinite obstruction in Q6 (Question 6; Proposition 11.1 below proves its

regularity half). The intake found no false claim in the source. The computations reported below were made by the write with its own program (not shipped), an exhaustive enumeration of all 65,814 neutral rules in dimensions 1–5 in exact rational arithmetic; they are observations about those dimensions only.

1. **Fixed-dimension prefix constants** (Q1). Proved:  $K(n, m, r) \leq 2^m/V(m, r)$  (Theorem 4.1), and the rules  $F_\ell$  show that the bound is the supremum over  $n$ . The write adds three facts with proofs. (a)  $K(n, m, r)$  is nondecreasing in  $n$ : a neutral rule on  $n$  coordinates, read on  $n + 1$  coordinates by ignoring the last, is neutral, with the same change distances and the same  $\alpha_f(m)$  for  $m \leq n$ ; so the supremum over  $n$  is a limit. (b)  $K(n, m, r) \leq 1/(1 - \rho_n^*(r))$ , with  $\rho_n^*$  given by (4), and equality holds for  $m = n$ : always  $\alpha_f(m) \leq 1$ , and  $\varepsilon_f(r) \geq 1 - \rho_n^*(r)$  by Theorem 2.1, which a maximizer attains; for  $m = n$  the prefix determines  $f$ , so  $\alpha_f(n) = 1$  for every neutral  $f$ . (c)  $K(n, 2, 1) = 4/3$  for  $n \geq 3$ : majority of three votes has  $\alpha(2) = 1$  (two equal votes decide) and  $\varepsilon(1) = 3/4$ , so (a) gives  $K(n, 2, 1) \geq 4/3 = 2^2/V(2, 1)$ . *Observation*: for every  $n \leq 5$  and  $1 \leq r \leq m \leq n$ ,

$$K(n, m, r) = \min \left\{ \frac{2^m}{V(m, r)}, \frac{1}{1 - \rho_n^*(r)} \right\},$$

so the inequality of Theorem 4.1 is strict in those dimensions except when  $r = m$  (both sides are 1) or  $(m, r) = (2, 1)$  with  $n \geq 3$ ; for example  $K(5, 3, 1) = 8/5 < 2$ ,  $K(5, 4, 2) = 16/15 < 16/11$  and  $K(5, 5, 1) = 8/5 < 16/3$ . Missing: whether the displayed minimum is  $K(n, m, r)$  for all  $n$  (the next dimension has  $2^{32}$  neutral rules), and in any case a formula for  $m < n$  and the rate at which  $K(n, m, r) \rightarrow 2^m/V(m, r)$ .

2. **Classification and stability of optimal rules** (Q2; the end of Section 2). Proved: majority on  $d(n)$  coordinates attains every budget at once. Maximizers are not unique: composing with an isometry of the cube (a permutation of coordinates and sign changes of some of them) preserves neutrality and every change distance, and for even  $n$  the ignored coordinate is arbitrary. For any maximizer of  $\rho_f(r)$  with nonempty core, Lemma 2.2 and (7) make  $C_r(f)$  a set of the largest size allowed by Theorem 2.3 for diameter  $n - 2r - 1$ , so the characterization of those extremal families (credited to Frankl in the abstract of [10]; not read by the write) describes the cores. The source's sketch stops there. Missing: control of the rule outside the two cores, which is what a classification of rules needs, and a stability statement for whole rules that is more than the known stability of cores.
3. **Two budgets** (Q3). Proved: Proposition 5.3 ( $\varepsilon(1) \rightarrow 0$  with  $\varepsilon(2) = 1$ ). *Observation*: the attainable pairs  $(\varepsilon_f(1), \varepsilon_f(2))$  are  $(1, 1)$  for  $n = 2$ ;  $(3/4, 1)$  and  $(1, 1)$  for  $n = 3$ ; three pairs, all with  $\varepsilon(2) = 1$ , for  $n = 4$ ; and  $(5/8, 15/16)$ ,  $(3/4, 1)$ ,  $(13/16, 1)$ ,  $(7/8, 1)$ ,  $(15/16, 1)$ ,  $(1, 1)$  for  $n = 5$ . So in dimensions up to five a two-vote guarantee exists only at the smallest one-vote vulnerability, where majority of five attains  $(5/8, 15/16)$ . Missing: the attainable region in general, and which restrictions (complexity, monotonicity, anonymity) force  $\varepsilon(2) \rightarrow 0$  from  $\varepsilon(1) \rightarrow 0$ .
4. **Dense extraction from majority** (Q4; the end of Section 6). Proved: the schedule (41) suffices. Since  $q_j \geq n_{j-1}$  and  $\delta_j^{-2} = 4^{j-1}/a^2 > 1$ , it forces  $n_j + 1 \geq 8n_{j-1}^2$ , so  $n_j > n_{j-1}^2$ : the schedule grows at least doubly exponentially. Missing: any lower bound on the growth needed for bounded density (35) or for equivalence with  $\beta$ , and the separation from pairwise decorrelation and from central limit theorems.
5. **A quantitative cost of changing one's mind** (Q5; the paragraph after Proposition 7.2). Proved: qualitative resilience gives no rate (Proposition 7.2). The source proposes assuming a bound such as  $\varepsilon_n(r) \leq Cr/\sqrt{n}$  and restricting ignored votes. Missing: any rate theorem.

6. **Several alternatives and other symmetry groups** (Q6). The source says the infinite obstruction “has a natural generalization”; Proposition 11.1 proves its regularity half for a finite alphabet and a group without a global fixed label. Missing: existence of such rules under choice (when an element of the group fixes an  $E_0$ -class the construction of Theorem 8.1 does not apply as it stands), the finite multi-part packing problem, and an analogue of the prefix inequality.
7. **Other adversarial metrics** (Q7). The source’s sketch separates two invariants: the smallest vulnerable fraction of a nonconstant fibre (Lemma 4.2 for Hamming distance) and the largest separated cores (Lemma 2.2). Missing: both invariants for any other metric.
8. **The exact weak-choice principle** (Q8). Proved: choice for the pairs of complementary  $E_0$ -classes, or a nonprincipal ultrafilter on  $\mathbb{N}$ , suffices. The write notes that the first is also necessary, over ZF: a rule  $F$  satisfying (46) is constant on each  $E_0$ -class and takes opposite values on complementary classes, so the classes on which  $F = 1$  select one class from each complementary pair. The question is therefore where this selection principle lies among weak choice principles (the sibling noisy-parity-cubes, np:found:choice, records the analogous equivalence for parity labels and a model of Serafin separating it from ultrafilters and  $E_0$ -transversals). Missing: that placement.
9. **Beyond independent biased inputs** (Q9). Theorem 9.1 uses the zero–one law for finite-change invariant sets and nonsingularity of finite flips. Missing: a criterion for dependent laws, or counterexamples isolating the hypothesis that fails.
10. **Effective and formal quantitative ergodicity** (Q10). Remark 10.1 shows what the existing Lean declarations give. The formalization order of Section 10 is a proposal. Missing: every formal statement of this report.

**Proposition 11.1** ([write] Several labels: the regularity obstruction, 5 October 2026). *Let  $\Sigma$  be a finite set with at least two elements, and let a group  $G$  act on  $\Sigma$  so that no element of  $\Sigma$  is fixed by all of  $G$ . Let  $G$  act on  $\Sigma^{\mathbb{N}}$  coordinatewise, and give  $\Sigma^{\mathbb{N}}$  the product topology and the uniform product measure  $\lambda$ . Suppose  $F : \Sigma^{\mathbb{N}} \rightarrow \Sigma$  satisfies  $F(x) = F(y)$  whenever  $x$  and  $y$  differ in finitely many coordinates, and  $F(gx) = gF(x)$  for all  $g \in G$  and all  $x$ . Then the fibres of  $F$  are not all measurable for the completion of  $\lambda$ , and they do not all have the Baire property. If  $G$  acts transitively on  $\Sigma$ , no fibre is measurable and no fibre has the Baire property.*

For  $\Sigma = \{-1, 1\}$  and  $G = \{\pm 1\}$  acting by multiplication this gives the nonmeasurability and the Baire-property statement of Theorem 8.1, without its inner- and outer-measure refinement.

*Proof.* Write  $A_s = F^{-1}(s)$ . For  $k \geq 1$  and a permutation  $\pi$  of  $\Sigma^k$ , let  $\phi_\pi$  replace the first  $k$  coordinates  $w$  of a point by  $\pi(w)$ . Each  $\phi_\pi$  is a homeomorphism preserving  $\lambda$  and changes only finitely many coordinates, so every fibre is invariant under all  $\phi_\pi$ ; for each  $k$  these maps act transitively on the cylinders  $[w]$ ,  $w \in \Sigma^k$ .

*Zero–one laws.* Let  $A$  be invariant under all  $\phi_\pi$ . If  $A$  is measurable for the completion, then  $\lambda(A \cap [w])$  is the same for all  $w \in \Sigma^k$ , so  $\lambda(A \cap [w]) = \lambda(A)\lambda([w])$ . The cylinders and  $\emptyset$  form a  $\pi$ -system generating the Borel sets, so  $A$  is independent of every Borel set; choosing a Borel  $B$  with  $\lambda(A \triangle B) = 0$  gives  $\lambda(A) = \lambda(A \cap B) = \lambda(A)^2$ , so  $\lambda(A) \in \{0, 1\}$ . If  $A$  has the Baire property and is not meager, it is comeager in some cylinder  $[w]$ ; each  $[w']$  with  $w'$  of the same length is the image of  $[w]$  under some  $\phi_\pi$ , which maps  $A$  to itself, so  $A$  is comeager in each of these finitely many cylinders, which cover  $\Sigma^{\mathbb{N}}$ , and  $A$  is comeager.

*The fibres.* For  $g \in G$  the map  $x \mapsto gx$  is a homeomorphism preserving  $\lambda$  (it permutes the letters of each coordinate), and it maps  $A_s$  onto  $A_{gs}$ , since  $F(gx) = gF(x)$ . If all fibres are measurable,

they are disjoint, cover the space and have measure 0 or 1, so exactly one,  $A_{s_0}$ , has measure one; then  $A_{gs_0} = gA_{s_0}$  has measure one, so  $gs_0 = s_0$  for every  $g$ , contrary to the hypothesis. If all fibres have the Baire property, each is meager or comeager; they are not all meager, since finitely many meager sets cannot cover the compact Hausdorff, hence Baire, space  $\Sigma^{\mathbb{N}}$ , and two disjoint sets cannot both be comeager; so exactly one,  $A_{s_0}$ , is comeager, and  $A_{gs_0} = gA_{s_0}$  is comeager, so again  $gs_0 = s_0$  for every  $g$ . If  $G$  acts transitively and one fibre  $A_s$  is measurable (or has the Baire property), then so is every  $A_{gs} = gA_s$ , that is, every fibre, and the previous case applies.  $\square$

## A A concise problem statement suitable for sharing

For every positive integer  $n$ , a committee rule receives  $n$  signs and returns one sign. Reversing every input sign must reverse its answer. Under uniform random input, call a configuration  $r$ -safe if no adversary changing at most  $r$  signs can change the answer.

Find the largest possible fraction of  $r$ -safe configurations, including the even/odd dependence. Then evaluate a neutral rule at each successive prefix of one infinite fair stream. Suppose, for every fixed  $r$ , the probability of being  $r$ -safe tends to one. Can the answers eventually settle with positive probability? Can they settle on a nonmeager set? Can a perfectly safe neutral answer be assigned directly to every infinite stream? Finally, determine whether one can select fixed committee sizes along which decisions behave statistically like an infinite fair-coin sequence while individual finite interventions eventually become harmless almost surely.

The finite answer is (4). Stabilization is impossible in both regularity senses. A direct everywhere assignment exists under choice and lacks both regularities. The selected-size statistical assertion holds in the precise bounded-density and coupling senses of Section 6.

## B Scope and assumptions at a glance

| Statement                     | Essential assumptions                                                             | Nature of the result                                                 |
|-------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Finite optimum                | Uniform finite cube; pointwise neutrality                                         | Corollary of Kleitman, with full reduction and matching construction |
| Sharp prefix inequality       | Uniform finite cube; pointwise neutrality                                         | Elementary proof and asymptotically matching examples                |
| No measurable limit           | Vanishing one-vote vulnerability, or vanishing influence of each fixed coordinate | Direct proof of a Rényi mixing consequence                           |
| Open near-Bernoulli recoding  | Vanishing finite-prefix conditional biases                                        | Deterministic subsequence, full-stream density comparison            |
| Diverging selected robustness | Vulnerability tends to zero for every fixed budget                                | Borel–Cantelli and finite-change invariance on a conull set          |
| Perfect infinite selector     | A sufficient choice principle                                                     | Exists globally; fair-product and Baire regularity impossible        |
| Biased-law threshold          | Independent nondegenerate coordinates; ZFC; completed measurability               | Corollary of Kakutani with a global choice extension                 |

If external randomness is introduced, neutrality should hold for each fixed seed. A distributionally neutral rule could simply use a single external fair coin forever and ignore the votes, which changes the problem. A mixture of fixed-seed neutral finite rules cannot exceed the deterministic finite optimum for the averaged robustness objective, because each seed already obeys that bound.

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